

SIAD HOSPITAL
SYNDROME OF INAPPROPRIATE ANTIDIURESIS

Uosm OVER 100 LOW URIC ACID EUVOLEMIC EQUALS SIAD

SIAD TREATMENT LADDER

NEW IN GUIDELINES
SGLT-2 INHIBITOR INDUCES GLYCOSURIA OSMOTIC DIURESIS RAISES Na

NEW
TOLVAPTAN 25mg V2 ANTAGONIST AQUARETIC

UREA 30 TO 60g PROMOTES FREE WATER EXCRETION QUESTIONABLE EFFICACY

SALT QUESTIONABLE EFFICACY

FLUID RESTRICT 500 TO 1000 mL PER DAY

SSRI

CARBAMAZEPINE

TCA

HYPERELECTROLYTES

CAUSES THIRST PLUS AVP

CHEMO

CHEMO INDUCED

DEMECCLOCYCLINE EUROPEAN GUIDELINES DO NOT RECOMMEND

ACUTE-RESCUE STATION
3 PERCENT SALINE 100 mL OVER 15 MIN. SEVERE NEUROLOGIC SYMPTOMS TARGET 4 TO 6Eq RISE

Uosm OVER 100
U Na OVER 30
URIC ACID LESS 4
EUVOLEMIC NORMAL VOLUME

SIAD

NO COPEPTIN X DIAGNOSIS
PLASMA AVP COPEPTIN HAS NO DIAGNOSTIC VALUE

PATIENTS:
SCLC 75 PERCENT OF MALIGNANCY
SIAD CAUSER
PULMONARY PNA TB ABSCESS
CNS MENINGITIS SAH & TRAUMA
CNS TRAUMA TUMOR
POSTOP NAUSEA
PAIN MEDICATION ABUSE
ENDURANCE EXERCISE
HYPOKALEMIA

1. Uosm >100, low uric acid, euvolemic

WHAT YOU SEE

Banner: Uosm over 100, low uric acid, euvolemic equals SIADH

WHAT IT ENCODES

SIADH (syndrome of inappropriate antidiuresis) is EUVOLEMIC hypotonic hyponatremia caused by AVP secretion that persists despite low plasma osmolality. The diagnostic triad: serum osm low (<275), urine inappropriately concentrated (Uosm >100), and urine Na high (>30–40) in a clinically euvolemic patient, supported by low serum uric acid (<4) and normal thyroid/adrenal axes. It is a diagnosis of exclusion.

BOARD TRAP

WRONG: SIADH causes hypoVOLEMIC hyponatremia. Discriminator — SIADH is EUVOLEMIC; the retained water distributes across total body water without edema or signs of depletion. Hypovolemic hyponatremia (vomiting, diuretics) has low urine Na (<20) when the cause is extrarenal — the opposite of SIADH's high urine Na.

2. Euvolemic / normal volume

WHAT YOU SEE

Hooded figure labeled euvolemic, normal volume

WHAT IT ENCODES

SIADH patients are clinically EUVOLEMIC — no edema, no JVD, no orthostasis, no signs of dehydration. Water retention triggers a transient natriuresis (pressure/volume-mediated) that prevents true volume overload, so the excess water is dilutional rather than third-spaced. Volume status on exam is the key bedside step in classifying hyponatremia.

BOARD TRAP

WRONG: SIADH presents with edema and volume overload. Discriminator — edematous hyponatremia is HYPERvolemic (CHF, cirrhosis, nephrotic syndrome) with low urine Na from neurohormonal avidity; SIADH is euvolemic with HIGH urine Na. Seeing edema should steer you away from SIADH.

3. Uosm over 100

WHAT YOU SEE

Sign: Uosm over 100

WHAT IT ENCODES

In SIADH urine osmolality is INAPPROPRIATELY high (>100, frequently >300) even though serum is hypotonic — the hallmark of unsuppressed AVP. A normally functioning kidney facing hypotonicity would maximally dilute the urine to <100, so a Uosm above 100 in a hypotonic patient signals continued ADH action.

BOARD TRAP

WRONG: the urine is maximally dilute (Uosm <100) in SIADH. Discriminator — maximally dilute urine in hypotonic hyponatremia points to primary polydipsia or low solute intake (beer potomania/tea-and-toast), NOT SIADH; SIADH's defining feature is FAILURE to dilute (Uosm >100).

4. Urine Na over 30

WHAT YOU SEE

Sign: U Na over 30

WHAT IT ENCODES

Urine sodium is high (>30–40 mmol/L) in SIADH because the patient is euvolemic — there is no hypovolemic stimulus for the kidney to avidly retain sodium, so it excretes Na normally. This separates SIADH from hypovolemic hyponatremia, where extrarenal losses drive urine Na <20.

BOARD TRAP

WRONG: urine sodium is low (<20) in SIADH. Discriminator — low urine Na (<20) indicates a volume-depleted/avid state (true hypovolemia, CHF, cirrhosis); SIADH's euvolemia leaves urine Na HIGH (>30). (Caveat: very low dietary salt intake can lower it.)

5. Uric acid less than 4

WHAT YOU SEE

Sign: uric acid less than 4

WHAT IT ENCODES

Low serum uric acid (<4 mg/dL) supports SIADH: mild volume expansion increases renal urate clearance (and dilution lowers concentration), so hypouricemia plus a high fractional excretion of urate is a useful confirmatory clue. Cerebral salt wasting can also lower uric acid, but it differs by being HYPOvolemic.

BOARD TRAP

WRONG: a HIGH serum uric acid supports SIADH. Discriminator — SIADH characteristically gives LOW uric acid from volume-expansion-driven urate clearance; a high uric acid suggests a volume-contracted state instead. Hypouricemia is one of the supporting clues, not hyperuricemia.

6. Low serum osmolality (hypotonic)

WHAT YOU SEE

No separate icon — read it off the top banner UOSM OVER 100, LOW URIC ACID, EUVOLEMIC: SIADH is the picture only once the plasma is confirmed truly hypotonic (low serum osm), the unstated precondition behind that diagnostic triad.

WHAT IT ENCODES

SIADH is a HYPOtonic (true) hyponatremia — measured serum osmolality is low (<275 mOsm/kg). Before labeling it SIADH you must exclude pseudohyponatremia (severe hyperlipidemia/hyperproteinemia, which gives a normal measured osm) and hypertonic hyponatremia from hyperglycemia or mannitol (high measured osm). Confirming low measured serum osm is step one.

BOARD TRAP

WRONG: SIADH has a high serum osmolality. Discriminator — high measured serum osm with hyponatremia indicates a hypertonic cause (hyperglycemia, mannitol) pulling water out of cells, not SIADH; SIADH is by definition HYPOtonic (low serum osm).

7. No copeptin diagnostic value

WHAT YOU SEE

Sign: no copeptin diagnosis, plasma AVP/copeptin no diagnostic value

WHAT IT ENCODES

Measuring plasma AVP or copeptin does NOT diagnose SIADH — levels overlap with other hyponatremias and add nothing, so SIADH remains a clinical/biochemical diagnosis of exclusion (euvolemia + Uosm >100 + UNa >30 + normal thyroid/adrenal/renal function). This is the opposite of DI, where copeptin is central to typing.

BOARD TRAP

WRONG: a high copeptin (or AVP) level confirms SIADH. Discriminator — AVP/copeptin can be elevated in many hyponatremias (volume depletion, CHF) and is not specific; SIADH is diagnosed by the euvolemic biochemical pattern and exclusion of other causes, not by an AVP level.

8. CNS causes

WHAT YOU SEE

Bed labeled CNS meningitis, SAH, trauma

WHAT IT ENCODES

SIADH causes — CNS disease: ischemic/hemorrhagic stroke, subarachnoid hemorrhage, meningitis/encephalitis, head trauma, brain tumor, and Guillain-Barré. These central insults inappropriately stimulate hypothalamic AVP release. SAH/intracranial disease also classically raises the differential of cerebral salt wasting, which must be distinguished by volume status.

BOARD TRAP

WRONG: CNS bleeds cause cerebral salt wasting only and never SIADH. Discriminator — intracranial disease causes BOTH; SIADH is euvolemic/hypervolemic (UNa high, treat with fluid restriction) while cerebral salt wasting is HYPOvolemic (also UNa high, but treat with salt and volume). Volume status, not urine Na, separates them.

9. Pulmonary causes

WHAT YOU SEE

Bed labeled pulmonary, pneumonia, TB, abscess

WHAT IT ENCODES

SIADH causes — pulmonary disease: pneumonia (especially Legionella), tuberculosis, lung abscess, and positive-pressure mechanical ventilation. Inflamed/diseased lung tissue and raised intrathoracic pressure stimulate AVP release; hyponatremia in a pneumonia patient often reflects SIADH and tends to resolve as the infection treats.

BOARD TRAP

WRONG: only malignant lung disease (not infection) causes SIADH. Discriminator — benign pulmonary infections (pneumonia, TB, abscess) are common SIADH triggers, separate from ectopic AVP production by small cell lung CANCER; both lung infection and lung cancer belong on the SIADH causes list.

10. SCLC / malignancy

WHAT YOU SEE

Bed labeled SCLC, cancer, malignancy

WHAT IT ENCODES

SIADH causes — malignancy: SMALL CELL lung carcinoma is the classic ectopic-AVP producer (a paraneoplastic syndrome); also head and neck cancers, and others. Persistent unexplained SIADH in a smoker should prompt evaluation for occult small cell lung cancer.

BOARD TRAP

WRONG: SQUAMOUS cell lung cancer is the classic SIADH tumor. Discriminator — SMALL cell lung cancer is the paraneoplastic ectopic-ADH producer (and also the classic ectopic-ACTH producer); squamous cell lung cancer is instead the classic cause of paraneoplastic HYPERcalcemia (PTHrP).

11. Drugs: SSRI / carbamazepine

WHAT YOU SEE

SSRI + carbamazepine pill boxes

WHAT IT ENCODES

SIADH causes — drugs: SSRIs (especially in the elderly), carbamazepine/oxcarbazepine, cyclophosphamide (and vincristine), antipsychotics, NSAIDs (potentiate AVP by blocking opposing prostaglandins), thiazides, and MDMA/ecstasy. Drug-induced SIADH usually improves on stopping the offending agent; review the med list in any new hyponatremia.

BOARD TRAP

WRONG: SSRIs and carbamazepine do not cause hyponatremia. Discriminator — both are well-known SIADH drugs (SSRIs notably in older patients); a new hyponatremia after starting an SSRI, carbamazepine, or a thiazide should trigger a med review before invasive workup.

12. Fluid restriction (first-line)

WHAT YOU SEE

Treatment ladder: fluid restrict

WHAT IT ENCODES

First-line treatment of chronic/asymptomatic SIADH is FLUID RESTRICTION (commonly <800–1000 mL/day, and less than urine output). If inadequate, add oral salt tablets and/or oral urea, or a loop diuretic. Treat the underlying cause and stop offending drugs. Restriction is less effective when urine osm is very high (>500) or UNa+UK exceeds serum Na.

BOARD TRAP

WRONG: first-line treatment for all SIADH is hypertonic saline. Discriminator — 3% saline is reserved for SEVERE symptomatic (seizure/coma) hyponatremia; routine chronic/asymptomatic SIADH is managed with fluid restriction ± salt/urea, not hypertonic saline.

13. Vaptans / urea

WHAT YOU SEE

Tolvaptan V2 antagonist, urea promotes water excretion

WHAT IT ENCODES

Second-line options promote free-water (aquaretic) excretion: vaptans — tolvaptan (oral) and conivaptan (IV) — are V2-receptor ANTAGONISTS, and oral urea induces an osmotic water diuresis. Tolvaptan requires in-hospital initiation, carries a hepatotoxicity warning, and must be started without concurrent fluid restriction to avoid over-rapid correction. Demeclocycline is older and largely avoided (nephrotoxicity).

BOARD TRAP

WRONG: vaptans (tolvaptan/conivaptan) are V1 receptor AGONISTS. Discriminator — vaptans are V2 ANTAGONISTS that block AVP's renal water-reabsorption action to produce a free-water aquaresis; calling them V1 agonists inverts both the receptor and the direction of effect.

14. 3% hypertonic saline (severe)

WHAT YOU SEE

Acute-rescue station: 3 percent saline 100 ml

WHAT IT ENCODES

For SEVERE symptomatic hyponatremia (seizures, obtundation, coma), give 3% hypertonic saline as a bolus (e.g., 100–150 mL, repeatable) targeting a prompt but limited rise of about 4–6 mmol/L to relieve cerebral edema. After the initial controlled bump, switch to careful management while respecting the 24-h correction ceiling.

BOARD TRAP

WRONG: give hypertonic saline rapidly to fully normalize the sodium. Discriminator — the goal is a small, controlled rise (~4–6 mmol/L) to break severe symptoms, NOT normalization; correcting too fast/too far risks osmotic demyelination. Hypertonic saline is rescue therapy, not a means to a normal Na in one sitting.

15. Correction rate limit / ODS

WHAT YOU SEE

Warning: severe neuro symptoms, target 4-6 mmol/L

WHAT IT ENCODES

Limit sodium correction to <8 mmol/L per 24 h (be even more cautious — <6–8/24h — in chronic hyponatremia or high-risk patients: alcoholism, malnutrition, liver disease, hypokalemia). Over-rapid correction causes OSMOTIC DEMYELINATION SYNDROME (central pontine myelinolysis): delayed dysarthria, dysphagia, quadriparesis, and locked-in syndrome. If correction overshoots, re-lower Na with D5W ± desmopressin.

BOARD TRAP

WRONG: chronic hyponatremia should be corrected as fast as possible. Discriminator — rapid correction of CHRONIC hyponatremia causes osmotic demyelination; keep the rise <8 mmol/L/24h. Only the initial relief of severe symptoms is brisk (~4–6 mmol/L), after which the daily ceiling governs.
